TITLE OF PROJECT

AI Mental Health Text Analyzer (CLI)

Submitted by

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Course: Fundamentals in AI and ML (CSA2001)

• TITLE

AI Mental Health Text Analyzer — A Command-Line Based Emotion Detection System

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• INTRODUCTION

Artificial Intelligence is increasingly being used to solve real-world problems in simple and effective ways. This project presents a Command Line Interface (CLI) based application that analyzes user input text and detects emotional state such as happiness, sadness, stress, or anger. Based on the detected emotion, the system provides meaningful suggestions to improve the user's mental well-being. The system is lightweight, easy to use, and does not require any graphical interface.

• MOTIVATION OF THE PROJECT

In today's fast-paced academic environment, students often experience stress, anxiety, and emotional pressure. However, not everyone has access to immediate help or guidance. This project was motivated by the need to create a simple, accessible tool that can assist users in understanding their emotions and provide basic suggestions through an AI-based approach.

• PROBLEM STATEMENT

There is a lack of simple and accessible tools that can analyze user emotions from text input and provide helpful suggestions in a lightweight, terminal-based environment.

• OBJECTIVE OF THE PROJECT

- To develop a CLI-based AI application
- To detect emotions from user input text
- To provide helpful suggestions based on emotions
- To demonstrate the use of AI concepts in a real-world scenario

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• EXISTING METHODS

- Online therapy platforms
- AI chatbots
- Mobile mental health applications

• PROS AND CONS OF THE STATED METHODS

Pros:

- Provide professional support
- Interactive and intelligent systems
- Widely available

Cons:

- Require internet connection
- Complex user interfaces
- Privacy concerns
- Not lightweight

• HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Laptop or Desktop Computer
- Minimum 4GB RAM

Software:

- Python 3.x
- Command Line Interface (Terminal / CMD)
- Code Editor (VS Code recommended)

• METHODOLOGY AND GOAL

The system is developed using Python and follows a modular approach. The user interacts with the system through a CLI menu. The input text is processed using simple Natural Language Processing (NLP) techniques such as keyword matching. Based on detected keywords, the system classifies the emotion and generates suggestions. Data is stored using JSON files for maintaining user history.

• FUNCTIONAL MODULES DESIGN AND ANALYSIS

1. **Input Module**
   Handles user input through CLI
2. **Text Processing Module**
   Converts input into analyzable format
3. **Emotion Detection Module**
   Identifies emotion using keyword-based logic
4. **Suggestion Module**
   Provides recommendations based on emotion
5. **Data Storage Module**
   Stores user history in JSON format

• ALGORITHM DEVELOPMENT

1. Start program
2. Display menu options
3. Take user input
4. Process text input
5. Detect emotion using keywords
6. Generate suggestion
7. Store result in file
8. Repeat until user exits

• SOFTWARE ARCHITECTURAL DIAGRAM

User Input (CLI)
↓
Main Program (Python)
↓
Emotion Detection Module
↓
Suggestion Engine
↓
Data Storage (JSON)
↓
Output (Terminal)

• CODING

(Refer to the attached Python source code in the GitHub repository)

• OUTPUT

Welcome to Mental Health AI Assistant

1. Analyze Text
2. View History
3. Exit

Enter your choice: 1
Enter your feelings: I am stressed about exams

Detected Emotion: Stress
Suggestion: Take breaks and plan your work.

• KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- CLI-based interaction
- Emotion detection using keyword matching
- Suggestion generation system
- JSON-based data storage
- Modular programming approach

• SIGNIFICANT PROJECT OUTCOMES

- Developed a working AI-based CLI application
- Understood real-world use of AI concepts
- Improved coding and logical thinking skills

• TESTING AND REFINEMENT

- Tested all menu options
- Verified correct emotion detection
- Handled invalid and empty inputs
- Improved user prompts and flow

• PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

- Student mental health tools

- Basic AI assistants
- CLI-based support systems
- Educational applications

• CONTRIBUTION / FINDINGS OF THE PROJECT

- Demonstrated how simple AI techniques can solve real problems
- Showed effectiveness of CLI applications
- Highlighted importance of accessible tools

• LIMITATION / CONSTRAINTS OF THE SYSTEM

- Uses rule-based logic instead of advanced ML
- Limited emotion detection accuracy
- No graphical interface

• CONCLUSIONS

The project successfully demonstrates the implementation of a CLI-based AI system for emotion detection and suggestion generation. It provides a simple yet effective solution to a real-world problem using basic AI concepts.

• FUTURE ENHANCEMENTS

- Integration with machine learning models
- Addition of voice input
- Development of GUI version
- Improved NLP techniques

• REFERENCES

- Python Official Documentation
- Course Materials (AI & ML)
- Online programming tutorials